Serial No: 122999

Registration No: CIIT/SP19-PEE-003/ISB

COMSATS University Islamabad

MUHAMMAD SHAFIQUE s/o INAYAT ULLAH

Of

Islamabad Campus

has been conferred upon the degree of

Doctor of Philosophy in Electrical Engineering

Given on this Third day of November two thousand and Twenty Five at Islamabad

Date of Issuance: 7th November 2025

Controller of Examinations

Rector

Registrar

COMSATS UNIVERSITY
ISLAMABAD

VERIFIED

017216

FAIZAN SHAFIQ
Deputy Controller of Examinations
COMSATS University Islamabad
(Principal Seat)

Prepared By: .....

Checked By: .....

Note: This degree is issued without alteration or erasure.

Contact Address:
COMSATS University Islamabad,
Park Road, Tarlai Kalan,
Islamabad, Pakistan.
Ph: +92-51-9247000-3
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Confidential

COMSATS University Islamabad

Formerly (COMSATS Institute of Information Technology)
Examinations Department (Principal Seat)
Park Road, Chak Shahzad, Islamabad, Pakistan. Tel: (92-51) 90495076, Fax: (92-51) 9247006
Email: exams.verification@comsats.edu.pk

No.CUI/CoE/Doc-Verif/ **V296/25**/839633 **November 20, 2025**

To Whom it May Concern

1. It is hereby confirmed that credentials issued to the following graduate is genuine and in order as per our record:-

Sr#	Graduate Name	CUI Reg. #	Document #	Ticket #
1	Muhammad Shafique	CIIT/SP19-PEE-003/ISB	Degree. # 122999	017216

* Formerly (COMSATS Institute of Information Technology)

2. The credentials were issued to the above-mentioned graduate, who was bona-fide student of COMSATS University Islamabad (CUI), Pakistan.
3. Copies of above-mentioned documents are hereby returned after verification.

(Faizan Shafiq)
Dy. Controller of Exams (PS)

Encl: **as above**

c.c 1. Controller of Exams;
2. Treasurer, CUI, along with following cheque(s)/deposit/bank e-mail in original:-

Sr. #	Cheque #	Date	Bank Name	Amount
1	Challan	20-11-2025	HBL	2000/-

3. Master File.

COMSATS University Islamabad

0105382

TRANSCRIPT

Date: 04-Nov-2025

Certified that MUHAMMAD SHAFIQUE s/o INAYAT ULLAH Registration No. CIIT/SP19-PEE-003/ISB Department of Electrical Engineering has completed and passed all the requisite courses / examinations for the degree of Doctor of Philosophy in Electrical Engineering on 24th September 2025 from Islamabad Campus.

The details of the courses passed are as follows:

Course Code	Semester / Course Title	Cr.	LG	CP	Course Code	Semester / Course Title	Cr.	LG	CP
	Spring 2019					**Fall 2022**			
ECI610 *	Advanced Digital Design	3	F	0.00	EEE900	PhD Thesis		IP	
ECI748	Machine Learning	3	B	9.30		**Spring 2023**			
EEE611	Stochastic Processes	3	A	10.65					
	Fall 2019				EEE900	PhD Thesis		IP	
ECI641	Digital Image Processing	3	A+	12.00		**Fall 2023**			
ECI743	Computer Vision	3	A	10.80					
					EEE900	PhD Thesis		IP	
	Spring 2020					**Spring 2024**			
ECI610	Advanced Digital Design *(Repeated)*	3	B	9.00					
ECI841	Advanced Topics in Image Processing	3	A+	12.00	EEE900	PhD Thesis		IP	
	Fall 2020					**Fall 2024**			
EEE900	PhD Thesis		IP		EEE900	PhD Thesis		IP	
	Spring 2021					**From 20 Jan 2025 to 24 Jul 2025**		Thesis Evaluation	
EEE900	PhD Thesis		IP			**Spring 2025**			
	Fall 2021				EEE900	PhD Thesis	9	Approved	
EEE900	PhD Thesis		IP						
	Spring 2022								
EEE900	PhD Thesis		IP						

Thesis Title: **Image Reconstruction Methods for Accelerated Dynamic MRI**

Controller of Examinations

Errors/Omissions Excepted

(Please Turn Over)

EXPLANATORY NOTES

1. The following explanations are to assist the reader with the interpretation of the transcript.
2. The Academic year normally comprises two regular semesters i.e. (16 weeks each) and summer session (8 weeks).
3. English is the medium of instruction.

4. **Legend**

Cr.	Credit of Courses
GP	Grade Point
L.G.	Letter Grade
NC	Non Credit
WD	Withdrawal
IP	In Progress

5. **Credit Value Definition**

A course of one credit represents one hour of lecture / three hours laboratory work per week, per semester.

6. **Marks**

c. Each course carries 100 marks of which 50% are reserved for mid term, quizzes, assignments, practicals & 50% marks for the terminal examinations.
d. Minimum pass marks in each course are 60%.

7. **Grading System**

(Marks %)	(Grade Points)	Letter Grade
90 & Above	4.00	A+
89	3.95	A
88	3.90	
87	3.85	
86	3.80	
85	3.75	
84	3.70	
83	3.65	
82	3.60	
81	3.55	
80	3.50	
79	3.45	B
78	3.40	
77	3.35	
76	3.30	
75	3.25	

(Marks %)	(Grade Points)	Letter Grade
74	3.20	B
73	3.15	
72	3.10	
71	3.05	
70	3.00	
69	2.95	C
68	2.90	
67	2.85	
66	2.80	
65	2.75	
64	2.70	
63	2.65	
62	2.60	
61	2.55	
60	2.50	
59 & Below	0.0	F

Note: This Transcript is issued without alteration or erasure.

COMSATS UNIVERSITY ISLAMABAD
VERIFIED
017190
FAIZAN SHAFIQ
Deputy Controller of Examinations
COMSATS University Islamabad
(Principal Seat)

Contact Address:

COMSATS University Islamabad,
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Email: exams.verification@comsats.edu.pk

No.CUI/CoE/Doc-Verif/ **V295/25/** 829370 **November 18, 2025**

To Whom it May Concern

1. It is hereby confirmed that credentials issued to the following graduate is genuine and in order as per our record:-

Sr#	Graduate Name	CUI Reg. #	Document #	Ticket #
1	Muhammad Shafique	CIIT/SP19-PEE-003/ISB	Trans. # 0105382	017190

* Formerly (COMSATS Institute of Information Technology)

2. The credentials were issued to the above-mentioned graduate, who was bona-fide student of COMSATS University Islamabad (CUI), Pakistan.
3. Copies of above-mentioned documents are hereby returned after verification.

(Faizan Shafiq)
Dy. Controller of Exams (PS)

Encl: **as above**

c.c 1. Controller of Exams;
2. Treasurer, CUI, along with following cheque(s)/deposit/bank e-mail in original:-

Sr. #	Cheque #	Date	Bank Name	Amount
1	Challan	18-11-2025	HBL	2000/-

3. Master File.

THE OHIO STATE UNIVERSITY

Rizwan Ahmad, Ph.D.
Associate Professor
Biomedical Engineering
Electrical and Computer Engineering
The Ohio State University

460 West 12th Ave, BRT 318
Columbus, OH 43210
Phone: 1 (614) 293-6251
Email: ahmad.46@osu.edu
https://u.osu.edu/ahmad/

March 14, 2024

Certification of Completion of One-year Fellowship

To whom it may concern,

I am pleased to recognize Mr. Muhammad Shafique for his commitment and research efforts during his one-year fellowship at The Ohio State University's Department of Biomedical Engineering and the Ohio State University Wexner Medical Center.

As a trainee visiting from COMSATS University Islamabad, Pakistan, from March 2023 to March 2024, Mr. Shafique diligently worked under my supervision to develop deep learning-based techniques aimed at accelerating MRI acquisition. His work has contributed valuable insights and has shown promising directions for future research in my lab.

Mr. Shafique's consistent efforts in mastering the complexities of machine learning algorithms have enhanced our ongoing projects and have offered a fresh perspective to our approach to MRI reconstruction, especially cardiac MRI performed under exercise stress. His work has been conducted with a dedication that speaks to his potential and growing skills in the field of medical imaging.

The Department of Biomedical Engineering acknowledges Mr. Muhammad Shafique's scholarly endeavors during his time at our institution. His academic growth and contributions have positively influenced the academic environment at The Ohio State University.

We believe the experience and knowledge gained during his fellowship will serve him well in his future academic and professional endeavors. It has been a pleasure to host Mr. Shafique at OSU, and we commend him for his dedication and hard work.

Best regards,

Rizwan Ahmad

Rizwan Ahmad

Certificate of Recognition

presented to

Muhammad Shafique

Visiting Fellow from COMSATS University Islamabad, Pakistan

for conducting supervised research in

Deep Learning-based Self-supervised Accelerated MRI

at

The Ohio State University

Cardiovascular Magnetic Resonance Imaging (CMR) Lab, Columbus OH, USA

March 2023 to March 2024

Rizwan Ahmad March 14, 2024

Dr. Rizwan Ahmad, Ph.D.

Associate Professor, Department of Biomedical Engineering

THE OHIO STATE UNIVERSITY